DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech. Branch : Computer Engineering/Computer Science and Engineering

Subject Code & Name: Database Systems(BTCOC501) Semester : V

Max Marks: 60 Date:06/02/2025 Duration: 3 Hr.

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

						(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**						**12**
1	Which type of data can be stored in the database?						1
		a. Image oriented data	b. Text, files containing data	c. Data in the form of audio or video	d. All of the above		
2	Database languages can be used to ________________ the data in the database.						1
		a. Read	b. Store	c. Update	d. All of the Mentioned.		
3	A database is _____ with Data Manipulation Language.						1
		a. Accessed and Multiplied	b. Added and Manipulated	c. Access and Manipulated	d. Added and Multiplied		
4	What is a relation in RDBMS?						1
		a. Key	b. Table	c. Row	d. Data Types		
5	The function that an entity plays in a relationship is called that entity's ______________						1
		a. Participation	b. Position	c. Role	d. Instance		
6	The descriptive property possessed by each entity set is __________						1
		a. Entity	b. Attribute	c. Relation	d. Model		

7	Database _____which is the logical design of the database, and the database ____which is a snapshot of the data in the database at a given instant in time.					1
	a. Instance, Schema	b. Relation, Schema	c. Relation, Domain	d. Schema, Instance		
8	The intersection operator is used to get the _____ tuples.					1
	a. Different	b. Common	c. All	d. Repeating		
9	The number of attributes in relation is called as its					1
	a. Cardinality	b. Degree	c. Tuples	d. Entity		
10	A unit of storage that can store one or more records in a hash file organization is denoted as					1
	a. Buckets	b. Disk pages	c. Blocks	d. Nodes		
11	A hash function must meet ________ criteria.					1
	a.Two	b. Three	c. Four	d. None of the Mentioned		
12	Which of the following makes the transaction permanent in the database?					1
	a. View	b. Commit	c. Rollback	d. Flashback		
Q. 2	**Solve the following.**					**12**
A)	Define an Entity and Attribute. Explain the different types of attributes that occur in ER diagram model with an example.					**6**
B)	Draw the symbols of ER diagram. Draw a diagram for library management system					**6**
Q.3	**Solve the following.**					**12**
A)	Define and differentiate the following relational algebra operators with suitable example 1. Cartesian product 2. Natural join					**6**
B)	What is relational algebra? Explain basic operations along with symbols of relational algebra.					**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	What is view what are its advantages explain views in SQL with example.					**6**

B)	What is normalization? Why we need to normalize the database tables.		**6**
C)	Write the SQL syntax for the following with the example 1. Select 2. Alter 3. Update		**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	Explain 1^{st},2^{nd} and3rd normal form with the help of suitable example		**6**
B)	What are the features of good relational Designs?		**6**
C)	Explain The Static Hashing and Dynamic Hashing.		**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	Define transaction and their properties with suitable example		**6**
B)	Explain ACID properties in details.		**6**
C)	Explain Time Stamp-Based Protocols.		**6**
	***** End *****		